# IMPORTANT INFORMATION ON THE SABER-C SPACER AND ANTERIOR CERVICAL PLATE

## DESCRIPTION

The SABER-C Spacer is a cervical interbody fusion device used to provide structural stability in skeletally mature individuals following discectomy. The SABER-C Spacer is inserted through an anterior cervical approach and is available in various geometric and material options to fit the clinical needs of the patient. The SABER-C Anterior Cervical Plate is an anterior cervical fixation device that is available in various geometric and fixation options and can be used with various types of interbody spacers.

The SABER-C Spacer is available as either a radiolucent polymer, with titanium plasma spray coating, or in an all-titanium alloy version, as specified in ASTM F2026, F3001, F136, and F1580. The SABER-C Anterior Cervical Plate has two fixation options, spikes or screws, and is manufactured from titanium alloy, as specified in ASTM F136 and F1472. The SABER-C Anterior Plate cannot be used with spikes when implanted without the Spacer component.

## INDICATIONS

The Elevation Spine Saber-C System is a cervical interbody fusion device intended for use in skeletally mature patients with degenerative disc disease (DDD) of the cervical spine (C2-T1) and is for use in multiple contiguous spinal levels. DDD is defined as discogenic pain with degeneration of the disc confirmed by history and radiographic studies. These patients should be skeletally mature and have had at least six (6) weeks of non-operative treatment. The SABER-C Spacer is to be filled with autograft bone and/or allogenic bone graft composed of cancellous and/or corticocancellous bone.

The SABER-C Spacer is intended to be used with supplemental fixation, such as anterior cervical plates or posterior cervical screw fixation. The SABER-C Spacer is available in a solid titanium and a porous 3D printed titanium offering. When the SABER-C Spacer is used with the SABER-C Anterior Cervical Plate and screws, the plate-spacer-screw assembly can be used as a stand-alone device with the SABER-C Anterior Cervical Plate acting as the supplemental fixation. When the SABER-C Spacer is used with the SABER-C Anterior Cervical Plate and spikes, the plate-spacer-spike assembly should be used with additional supplemental fixation such as posterior cervical screw fixation.

The SABER-C Anterior Cervical Plate, when used without the spacer component and with screws, is intended for anterior screw fixation to the cervical spine (C2-C7) for the following indications: degenerative disc disease (as defined by neck pain of discogenic origin with degeneration of the disc confirmed by patient history and radiographic studies), trauma (including fractures), tumors, deformity (defined as kyphosis, lordosis, or scoliosis), pseudoarthrosis, failed previous fusion, spondylolisthesis, and spinal stenosis. The Anterior Cervical Plate is not to be used with spikes when used without the Spacer component.

## WARNINGS

One of the potential risks identified with this system is death. Other potential risks which may require additional surgery, include:

- device component fracture,
- loss of fixation,
- non-union,
- fracture of the vertebrae,
- neurological injury, and
- vascular or visceral injury.

SABER-C Anterior Cervical Plates are not approved for screw attachment or fixation to the posterior elements (pedicles) of the cervical, thoracic, or lumbar spine. Plate contouring is not recommended.

Interbody fusion devices for the treatment of degenerative conditions are designed to withstand both full load bearing and the loads associated with long-term use which could result from the presence of non-union or delayed union.

Certain degenerative diseases or underlying physiological conditions such as diabetes, rheumatoid arthritis, or osteoporosis may alter the healing process, thereby increasing the risk of implant breakage or spinal fracture.

Patients with previous spinal surgery at the level(s) to be treated may have different clinical outcomes compared to those without previous surgery.

The components of this system are manufactured from PEEK radiolucent polymer, titanium alloy and commercially pure titanium. Mixing of stainless steel implant components with different materials is not recommended for metallurgical, mechanical and functional reasons.

These warnings do not include all adverse effects that could occur with surgery in general, but are important considerations to orthopedic implants. General surgical risks should be explained to the patient prior to surgery.

Use this device as supplied and in accordance with the handling and use information provided below.

## PRECAUTIONS

The implantation of cervical fixation and intervertebral fusion devices should be performed only by experienced spinal surgeons with specific training in the use of this system because this is a technically demanding procedure presenting a risk of serious injury to the patient. Preoperative planning and patient anatomy should be considered when selecting implant size.

Surgical implants must never be reused. An explanted implant must never be re-implanted. Even though the device appears undamaged, it may have small defects and internal stress patterns which could lead to breakage.

Adequately instruct the patient. Mental or physical impairment which compromises or prevents a patient's ability to comply with necessary limitations or precautions may place that patient at a risk during postoperative rehabilitation.

The SABER-C Spacer and SABER-C Anterior Cervical Plate have not been evaluated for safety and compatibility in the Magnetic Resonance ("MR") environment. These devices have not been tested for heating, migration or image artifact in the MR environment. Scanning a patient who has these devices implanted may result in patient injury.

## CONTRAINDICATIONS

Use of these implants is contraindicated in patients with the following conditions:

- Active systemic infection, infection localized to the site of the proposed implantation, or when the patient has demonstrated allergy or foreign body sensitivity to any of the implant materials.
- Signs of local inflammation.
- Prior fusion at the level(s) to be treated.
- Severe osteoporosis, which may prevent adequate fixation.
- Conditions that may place excessive stresses on bone and implants, such as severe obesity or degenerative diseases, are relative contraindications. The decision whether to use these devices in such conditions must be made by the physician considering the risks versus the benefits to the patient.
- Patients whose activity, mental capacity, mental illness, alcoholism, drug abuse, occupation, or lifestyle may interfere with their ability to follow postoperative restrictions and who may place undue stresses on the implant during bony healing and may be at a higher risk of implant failure.
- Any patient not willing to cooperate with postoperative instructions.
- Any condition not described in the indications for use.
- Fever or leukocytosis.
- Patient has known metal sensitivity.
- Pregnancy.
- Any other condition that would preclude the potential benefit of spinal implant surgery, such as the presence of tumors or congenital abnormalities, fracture local to the operating site, elevation of sedimentation rate unexplained by other diseases, elevations of the white blood count (WBC), or a marked left shift in the WBC differential count.
- Any case not needing a fusion.
- Patients with a known hereditary or acquired bone friability or calcification problem should not be considered for this type of surgery.
- These devices must not be used for pediatric cases or where the patient still has general skeletal growth.
- Any case where the implant components selected for used would be too large or too small to achieve a successful result.
- Any case that requires the mixing of metals from two different components or systems.
- Any patient having inadequate tissue coverage at the operative site or inadequate bone stock or quality.
- Any patient in which implant utilization would interfere with anatomical structures or expected physiological performance.

## COMPLICATIONS AND POSSIBLE ADVERSE EVENTS

Prior to surgery, patients should be made aware of the following possible adverse effects in addition to the potential need for additional surgery to correct these effects:

- Loosening, bending or breakage of components
- Displacement/migration of device components
- Tissue sensitivity to implant material
- Potential for skin breakdown and/or wound complications
- Non-union or delayed union or mal-union
- Infection
- Nerve damage, including loss of neurological function (sensory and/or motor), paralysis, dysesthesia, hyperesthesia, paresthesia, radiculopathy, reflex deficit, cauda equina syndrome
- Dural tears, cerebral spinal fluid leakage
- Fracture of vertebrae
- Foreign body reaction (allergic) to components or debris
- Vascular or visceral injury
- Change in spinal curvature, loss of correction, height and/or reduction
- Urinary retention or loss of bladder control or other types of disorders of the urogenital system
- Ileus, gastritis, bowel obstruction or other types of gastrointestinal system compromise
- Reproductive system compromise including impotence, sterility, loss of consortium and sexual dysfunction
- Pain or discomfort
- Bursitis
- Decrease in bone density due to stress shielding
- Loss of bone or fracture of bone above or below the level of surgery
- Bone graft donor site pain, fracture, and/or delayed wound healing
- Restriction of activities
- Lack of effective treatment of symptoms for which surgery was intended
- Need for additional surgical intervention
- Death

## PACKAGING

These implants and instruments may be supplied pre-packaged and sterile, using gamma irradiation. The integrity of the sterile packaging should be checked to ensure that sterility of the contents is not compromised. Packaging should be carefully checked for completeness and all components should be carefully checked to ensure that there is no damage prior to use. Damaged packages or products should not be used, and should be returned to Elevation Spine. During surgery, after the correct size has been determined, remove the products from the packaging using aseptic technique.

The instrument sets are provided nonsterile and are steam sterilized prior to use, as described in the STERILIZATION section below. Following use or exposure to soil, instruments and instrument trays and cases must be cleaned, as described in the CLEANING section below.

## HANDLING AND USE

All instruments and implants should be treated with care. Improper use or handling may lead to damage and/or possible malfunction. Products should be checked to ensure that they are in working order prior to surgery. All products should be inspected prior to use to ensure that there is no unacceptable deterioration such as corrosion (i.e. rust, pitting), discoloration, excessive scratches, notches, debris, residue, flaking, wear, cracks, cracked seals, etc. Non-working or damaged instruments should not be used, and should be returned to Elevation Spine.

Implants are single use devices and should not be cleaned. Re-cleaning of single use implants might lead to mechanical failure and/or material degradation. Discard any implants that may have been accidently contaminated.

## CLEANING

Instruments should be thoroughly cleaned separately from instrument trays and cases. Lids should be removed from cases for the cleaning process, if applicable. All instruments that can be disassembled must be disassembled for cleaning. All handles must be detached. Instruments may be reassembled following sterilization. The products should be cleaned using neutral cleaners before sterilization and introduction into a sterile surgical field or (if applicable) return of the product to Elevation Spine.

Cleaning and disinfecting can be performed with aldehyde-free solvents at higher temperatures. Cleaning and decontamination must include the use of neutral cleaners followed by a deionized water rinse. Note: certain cleaning solutions such as those containing formalin, glutaraldehyde, bleach and/or other alkaline cleaners may damage some devices, particularly instruments; these solutions should not be used.

The following cleaning methods should be observed when cleaning instruments and instrument trays and cases after use or exposure to soil, and prior to sterilization:

1. Immediately following use, ensure that the instruments are wiped down to remove all visible soil and kept from drying by submerging or covering with a wet towel.
2. Disassemble all instruments that can be disassembled.
3. Rinse the instruments under running tap water to remove all visible soil. Flush the lumens a minimum of 3 times, until the lumens flush clean.
4. Prepare Enzol® ® (or a similar enzymatic detergent) per manufacturer's recommendations.
5. Immerse the instruments in the detergent and allow them to soak for a minimum of 2 minutes.
6. Use a soft bristled brush to thoroughly clean the instruments. Use a pipe cleaner for any lumens. Pay close attention to hard to reach areas.
7. Using a sterile syringe, draw up the enzymatic detergent solution. Flush any lumens and hard to reach areas until no soil is seen exiting the area.
8. Remove the instruments from the detergent and rinse them in running warm tap water.
9. Prepare Enzol® ® (or a similar enzymatic detergent) per manufacturer's recommendations in an ultrasonic cleaner.
10. Completely immerse the instruments in the ultrasonic cleaner and ensure detergent is in lumens by flushing the lumens. Sonicate for a minimum of 3 minutes.
11. Remove the instruments from the detergent and rinse them in running deionized water or reverse osmosis water for a minimum of 2 minutes.
12. Dry instruments using a clean soft cloth and filtered pressurized air.
13. Visually inspect each instrument for visible soil. If visible soil is present, then repeat cleaning process starting with Step 3.

### Instrument Case/Tray Automated Cleaning Procedure

The following steps should be completed in sequence:

1. Prepare a neutral pH enzymatic cleaning solution using lukewarm water as per the manufacturer's recommendation.

2. Fully immerse the devices and allow to soak for a minimum of two (2) minutes.

3. Following the soak time, flush any lumens of the device using a syringe.

4. Rinse the devices under lukewarm running water for a minimum of one (1) minute, while agitating the devices. Agitation includes actuating all movable parts, such as opening and closing hinges and moving the devices around under the running water.

5. Use a clean soft bristled brush and/or pipe cleaner to brush and aid in the rinse for the exterior and interior of device components. Use a syringe to flush any lumens.

6. Place the devices back into the designated locations of the case/tray and load the case/tray set into an automated washer (Getinge 86-Series, Steris 444, or equivalent).

7. The washer cycle parameters are as follows:

| Phase | Recirculation Time (min.) | Water Temperature | Detergent |
|---|---|---|---|
| Pre-Wash 1 | 2:00 | Cold Water | N/A |
| Wash 1 | 4:00 | Hot Water | Enzymatic Cleaner |
| Wash 2 | 3:00 | 60 (140F) | Mild Detergent |
| Rinse 1 | 2:00 | Hot Water | N/A |
| Dry | 7:00 | N/A | N/A |

a) Utilize OptiPro® or Enzol® enzymatic cleaner or equivalent.

b) Utilize Prolystica® 2X mild detergent or equivalent.

c) Only purified water that has been filtered using deionization, reverse-osmosis, or ultra-filtration should be used for all final rinsing.

d) Visually examine to determine if all adherent visible soil has been removed. Repeat the cleaning procedure if visible debris is detected.

## CONTACT INFORMATION

Elevation Spine may be contacted at 1 (844) 415-0226. A surgical technique manual may be obtained by contacting Elevation Spine.

## STERILIZATION

SABER-C implants may be available sterile or nonsterile. Instruments are provided nonsterile. Nonsterile implants and instruments are to be sterilized by the end-user per the parameters below.

Sterile implants are sterilized by gamma radiation, validated to ensure a Sterility Assurance Level (SAL) of 10-6. Sterile products are packaged in a heat-sealed double pouch or container/pouch. The expiration date is provided on the package label. These products are considered sterile unless the packaging has been opened or damaged. Sterile implants that become nonsterile or have expired packaging are considered nonsterile and may be sterilized according to instructions for nonsterile implants and instruments below.

Nonsterile implants and instruments have been validated to ensure an SAL of 10-6. The use of an FDA-cleared wrap is recommended, per the Association for the Advancement of Medical Instrumentation (AAMI) ST79, Comprehensive Guide to Steam Sterilization and Sterility Assurance in Health Care Facilities. It is the end user's responsibility to use only sterilizers and accessories (such as sterilization wraps, sterilization pouches, chemical indicators, biological indicators, and sterilization cassettes) that have been cleared by the FDA for the selected sterilization cycle specifications (time and temperature).

When using a rigid sterilization container, the following must be taken into consideration for proper sterilization of Elevation Spine devices and loaded graphic cases:

- Recommended sterilization parameters are listed in the table below.
- Only FDA-cleared rigid sterilization containers for use with pre-vacuum steam sterilization may be used.
- When selecting a rigid sterilization container, it must have a minimum filter area of 176 in2 total, or a minimum of four (4) 7.5in diameter filters.
- No more than one (1) loaded graphic case or its contents can be placed directly into a rigid sterilization container.
- Stand-alone modules/racks or single devices must be placed, without stacking, in a container basket to ensure optimal ventilation.
- The rigid sterilization container manufacturer's instructions for use are to be followed; if questions arise, contact the manufacturer of the specific container for guidance.
- Refer to AAMI ST79 for additional information concerning the use of rigid sterilization containers.

For implants and instruments provided NONSTERILE, sterilization is recommended (wrapped or containerized) as follows:

| Pre-Vacuum |
|---|
| Temperature 132º C (270º F) |
| Cycle time 4 minutes |
| Minimum dry time 60 minutes |
| Minimum cool-down time outside chamber on wire rack 30 minutes |
| Double wrapped |

*These parameters are validated to sterilize only this device. If other products are added to the sterilizer, the recommended parameters are not valid and new cycle parameters must be established by the user. The autoclave must be properly installed, maintained, and calibrated. Ongoing testing must be performed to confirm inactivation of all forms of viable microorganisms.*

**CAUTION:** Federal (USA) Law Restricts this Device to Sale by or on the order of a Physician.

| | | | |
|---|---|---|---|
| REF | CATALOGE NUMBER | STERILE R | STERILIZED BY IRRADIATION |
| LOT | LOT NUMBER | | MANUFACTURER |
| | CAUTION | | USE BY (YYYY-MM-DD) |
| | SINGLE USE ONLY | | |

ELEVATION
SPINE

2511 Garden Rd. Suite B125
Monterey, CA 93940

LBL-SC0119 Rev E

LBL-SC0119 Rev E